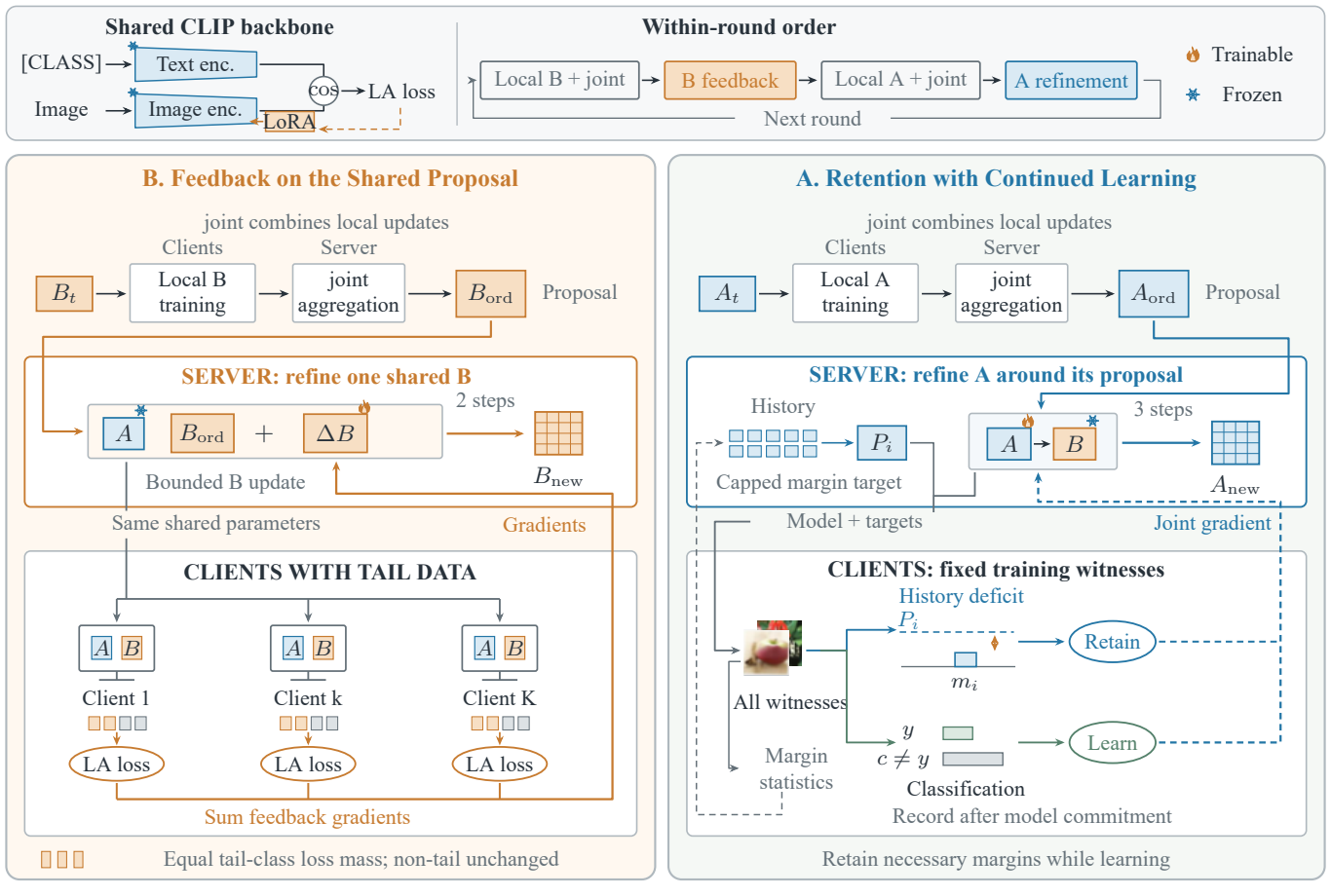

Shared CLIP backbone
[CLASS]
Text enc.
Image
Image enc.
LoRA
cos
LA loss
Within-round order
Local B + joint
B feedback
Local A + joint
A refinement
Next round
Trainable
Frozen
B. Feedback on the Shared Proposal
joint combines local updates
Clients
Server
B_t
Local B training
joint aggregation
B_{ord}
Proposal
SERVER: refine one shared B
2 steps
A
B_{ord}
+
ΔB
Bounded B update
B_{new}
Same shared parameters
Gradients
CLIENTS WITH TAIL DATA
Client 1
Client k
Client K
LA loss
LA loss
LA loss
Sum feedback gradients
Equal tail-class loss mass; non-tail unchanged
A. Retention with Continued Learning
joint combines local updates
Clients
Server
A_t
Local A training
joint aggregation
A_{ord}
Proposal
SERVER: refine A around its proposal
History
P_i
3 steps
A
B
Capped margin target
A_{new}
Model + targets
Joint gradient
CLIENTS: fixed training witnesses
History deficit
P_i
Retain
m_i
All witnesses
y
$c \neq y$
Learn
Margin statistics
Classification
Record after model commitment
Retain necessary margins while learning